Degree Project in Technology

Second cycle, 30 credits

Convolutional Auto Encoder for Generating Radar Range-Azimuth Maps from Images

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Master's Programme, Computer Science, 120 credits
Date: October 18, 2024

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Swedish title: Convolutional Auto Encoder för Genererar Radar Range-Azimuth Kartor från bilder

Abstract

The Society of Automotive Engineers has established a framework for driving automation, comprising six levels of autonomy, ranging from Level 0 (no driving automation) to Level 5 (full driving automation). To achieve the highest level of autonomy, vehicles must be equipped with sensors that enable them to accurately perceive their surroundings and make informed decisions. The ability to perceive the environment is crucial for safe and reliable vehicle operation, as it allows vehicles to detect and respond to various objects, obstacles, and road conditions. Effective perception requires the processing of complex sensor data, which can be achieved through the integration of multiple sensor modalities, including cameras, radar, and Light Detection and Ranging (LiDAR). Deep Learning has emerged as the preferred method for processing the vast amounts of data generated by these sensor modalities, due to its ability to learn complex patterns and relationships.

The development of deep learning-based assisted and automated driving technologies relies heavily on the availability of diverse and high-quality sensor data. However, the scarcity of publicly available radar datasets hinders the advancement of these technologies in the radar domain. To address this challenge, we propose Range Azimuth from Image, Depth and Semantic descriptors (RAIDS), a novel Convolutional Auto Encoder (CAE) based model that takes as input the Red Green Blue (RGB) image, depth information, and semantic scene descriptor obtained from two foundation models and is capable of converting them into Range-Azimuth (RA) maps. Notably, our model leverages attention mechanisms, specifically channel and spatial attention, to selectively focus on relevant features and improve the localization and estimation of reflection strength in the generated RA maps. To provide a comparison, we have also developed a model that uses only RGB images as input, without any additional depth or semantic descriptors, to synthesize RA maps.

The proposed model is trained using a publicly available dataset which contains scenes from urban, rural and highway environments using a long range radar to guide the learning process. Notably, our model demonstrates effective transfer learning capabilities, as it is able to generalize well to a different dataset, which contains scenes with different semantic classes and is captured with a short-range radar. This evaluation provides a robust test of the model's ability to adapt to new environments and different radar configurations. An ablation study is done to show the added benefit of each of supplementary information from different foundation models. Quantitative conclusion are drawn from a series of evaluation metrics and qualitative conclusions are drawn using diverse scenes from the dataset.

Keywords

Automotive radar, Semantic segmentation, Depth map estimation, Foundation models, Deep autoencoders.

Sammanfattning

Society of Automotive Engineers har etablerat ett ramverk för körning automation, som omfattar sex nivåer av autonomi, från nivå 0 (nr körautomation) till nivå 5 (full körautomation). För att uppnå högsta nivå av autonomi, fordon måste vara utrustade med sensorer som möjliggör dem att korrekt uppfatta sin omgivning och fatta välgrundade beslut. Förmågan att uppfatta miljön är avgörande för ett säkert och pålitligt fordon drift, eftersom den tillåter fordon att upptäcka och reagera på olika föremål, hinder och vägförhållanden. Effektiv uppfattning kräver bearbetning av komplexa sensordata, som kan uppnås genom integrering av flera sensormodaliteter, inklusive kameror, radar och lidar. Deep Learning har framkommit som den föredragna metoden för att bearbeta stora mängder data genereras av dessa sensormodaliteter, på grund av dess förmåga att lära sig komplex mönster och relationer.

Utvecklingen av Deep Learning-baserad assisterad och automatiserad körning teknologier är starkt beroende av tillgången på olika och högkvalitativa sensordata. Men bristen på allmänt tillgängliga radaruppsättningar hindrar utvecklingen av dessa tekniker. För att möta denna utmaning föreslår vi RAIDS, en ny Convolutional Auto Encoder (CAE) baserad modell som tar som inmatning av RGB-bilden, djupinformation och semantisk scenbeskrivning erhålls från två grundmodeller och kan konvertera dem i Range-Azimut (RA) kartor. Det är särskilt viktigt att vår modell drar nytta av uppmärksamhet mekanismer, särskilt kanal och rumslig uppmärksamhet, för att selektivt fokusera på relevanta egenskaper och förbättra lokaliseringen och uppskattningen av reflektion styrka i de genererade RA-kartorna. För att ge en jämförelse har vi också utvecklat en modell som endast använder rödgrönblå (RGB) bilder som indata, utan ytterligare djup eller semantiska deskriptorer, för att syntetisera RA-kartor.

Den föreslagna modellen är tränad med scener från stad, landsbygd och motorväg miljöer som använder en långdistansradar för att styra inlärningsprocessen. De Modellen visar också generaliseringsförmåga när den exponeras för scen från olika miljöer och även testad för en kortdistansradar. En ablation studien görs för att visa den extra fördelen med var och en av kompletterande information från olika grundmodeller. Kvantitativa slutsatser dras från a serier av utvärderingsmått och kvalitativa slutsatser dras med hjälp av olika scener från datamängden.

Nyckelord

Fordonsradar, Semantisk segmentering, Uppskattning av djupkarta, Foundation modeller, Deep autoencoders.

Acknowledgments

I would like to thank my supervisors at NXP, Dr. Ashish Pandharipande and Dr. Tunc Alkanat, for providing this opportunity and bearing with me while I ask simple questions about the workings of radar. I would also like to thank my supervisor at KTH, Dr. Patric Jensfelt for his constant feedback through a fresh set of eyes and providing invaluable guidance with the writing of this thesis. At this point, I have a heartfelt appreciation for all my friends from university and the ones I made along the way at NXP for making the whole journey better. Finally, I probably owe a lot of this thesis to Michael Bublé, "Feeling Good" was a banger way to start off the mornings.

Eindhoven, The Netherlands, October 2024
Pavan Aakash Rangaraj

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List of acronyms and abbreviations

ℓ_1 loss	Mean Absolute Error
CAE	Convolutional Auto Encoder
CBAM	Convolutional Block Attention Module
CGAN	Conditional Generative Adversarial Network
EM	Electro Magnetic
FMCW	Frequency-Modulated Continuous-Wave
GAN	Generative Adversarial Network
LiDAR	Light Detection and Ranging
MDE	Monocular Depth Estimation
ML	Machine Learning
MMDE	Monocular Metric Depth Estimation
MS-SSIM	Multi-Scale Structural Similarity Index Metric
MSE	Mean Squared Error
PSNR	Peak Signal to Noise Ratio
RA	Range-Azimuth
RAIDS	Range Azimuth from Image, Depth and Semantic descriptors
RD	Range-Doppler
RGB	Red Green Blue
SSIM	Structural Similarity Index Metric

Chapter 1

Introduction

The rapid advancement of assisted and automated driving technologies has led to an increased demand for robust and accurate scene understanding systems. Various sensor modalities are employed, each with their strengths: cameras provide rich semantic information, Light Detection and Ranging (LiDAR) excels in depth estimation, and radar offers valuable Doppler (velocity) and depth information, as well as the ability to operate effectively in adverse weather conditions where cameras may be hindered. Machine Learning (ML) methods have been recently used in different parts of the radar processing chain for enhanced detection and perception (1; 2). However, the limited amount of publicly available automotive radar datasets is hampering the development of such ML algorithms (3; 4), in particular deep learning approaches which rely on large amounts of radar data for model training. Meanwhile, camera images, both real-world and synthetic, are more abundant and easily accessible, driven by their widespread adoption in applications such as scene understanding (5) and mapping (6), with numerous datasets available for research purposes (7; 5).

While generating synthetic radar data from camera images is a promising approach, it remains a relatively unexplored area of research, with limited existing work in this domain (4). Our aim is to generate synthetic radar data from camera images, a challenging cross-modality conversion problem. This problem is closely related to other cross-modal conversion tasks, such as LiDAR to radar (8), Image-to-Point Cloud (4), and cross-modal fusion of image and LiDAR (9). These works have demonstrated the potential of cross-modal learning and fusion for various applications, including autonomous driving, robotics, and synthetic data generation. Building upon these ideas, our work aims to bridge the gap in the literature by exploring the possibilities and challenges of generating synthetic radar data from camera images. In this work, we concentrate on the synthesis of Range-Azimuth (RA) maps. A RA map is a two-dimensional representation that visualizes the reflection strength of targets in a scene, with the range of the targets relative to the radar position depicted on the vertical axis (y-axis) and their azimuth angle relative to the radar orientation plotted on the horizontal axis (x-axis). This map provides a spatially organized display of the radar returns, allowing for the identification of target locations and their corresponding reflection intensities with respect to the radar viewpoint. We propose a novel approach that leverages dense depth maps and semantic class information from images as supplementary sources of information to enable a deep learning model to learn deep patterns and transfer a Red Green Blue (RGB) image to a radar RA map.

1.1 Ethics and Sustainability

The research presented in this thesis has the potential to enhance the efficiency and sustainability of obtaining radar data in the field of assisted and autonomous driving. By developing a deep learning model capable of converting publicly available driving images into synthetic radar data, we can substantially reduce the need for manual data collection. This reduction in manual data collection efforts translates to significant time savings. For instance, the dataset used in this research, which contains sequences of two hours in length, can be converted into synthetic radar data in less than eight minutes using our model. This represents a substantial reduction in data collection time, from hours to mere minutes.

In addition to the time savings, our approach also addresses the issue of large space requirements for radar data. The raw output from the radar, which is required to build the RA map, is a three-dimensional array of size $L \times K \times N$, where L is the chirp rate of the radar which is the number of pulses the radar sends out in a time-period, K is samples per chirp which is the number of samples from a single pulse at one receiving antenna and N is the number of receiving antennas. For the training dataset we use in this research this corresponds to $512 \times 256 \times 192$. This radar cube is around 12 times larger than a RGB image stored at full 1080×1920 resolution. This raises both a space and computational issue, which hinders the collection of long sequences of data. By using a deep learning model, we can bypass the complex processing and large intermediate results that require large storage requirements.

The environmental benefits of this approach are equally noteworthy. Traditional data collection methods using radar-equipped vehicles result in greenhouse gas emissions, primarily carbon dioxide, which contribute to climate change. By minimizing the need for physical data collection, our method can help decrease the carbon emissions associated with this sector by utilising existing driving camera images available on the Internet.

1.2 Problem

Our proposed method aims to learn a transfer function from the camera image domain to the radar RA domain. Given an input image $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$, where H and W denote the height and width of the camera image respectively, and C is the number of channels, we concatenate the input image with depth information $\mathbf{D} \in \mathbb{R}^{H \times W}$ and semantic scene descriptors $\mathbf{S} \in \mathbb{R}^{H \times W}$, which have the same spatial dimensions as the input image $\mathbf{I}$. The depth map and semantic map are generated by two foundation models, namely UniDepth (10) and Mask2Former (11). This supplementary information enables our model to leverage explicit depth and semantic cues, thereby bypassing the need to infer them from the input images. Instead, our model can focus on learning the complex mapping between the RGB image and RA domain. Furthermore, by using foundation models to obtain the supplementary information, our models can tap into the collective knowledge and patterns that were learnt from the large datasets used to train the foundation models. The main objective of our proposed model is to determine a mapping f,

$$\mathbf{R} = f(\mathbf{I}, \mathbf{D}, \mathbf{S}; \theta), \tag{1.1}$$

where $\mathbf{R} \in \mathbb{R}^{H \times W}$ is the RA map corresponding to the inputs $\mathbf{I}, \mathbf{D}$ and $\mathbf{S}$. $\mathbf{R}$ has the same height H and width W as $\mathbf{I}$, and θ represents the learnable weights that parameterize the transfer function, allowing the model to adapt to the specific mapping between the camera image and RA domains.

1.3 Research Methodology

To comprehensively evaluate the performance of the proposed methods, we employ a mixed-methods approach, combining both quantitative and qualitative analysis. As a first step, an ablation study is conducted to investigate the contribution of various input strategies. This involves systematically removing and adding different scene descriptors to elucidate their impact on model performance. The results of this study provide valuable insights into the importance of incorporating additional scene information and the resultant performance gains.

Based on the findings of the ablation study, the best-performing model is selected for further evaluation. This model is then subjected to a comprehensive evaluation using a suite of metrics commonly used in image generation and image reconstruction tasks. Specifically, pixel-wise metrics such as Mean Squared Error (MSE) and Mean Absolute Error (ℓ_1 loss) are utilized to assess the accuracy of the generated images. Additionally, the Structural Similarity Index Metric (SSIM) is employed as a perceptual metric to evaluate the visual quality of the generated images. We provide a more detailed discussion on the SSIM metric in section 2.3. Furthermore, leveraging the radar domain, Peak Signal to Noise Ratio (PSNR) is used to assess the distortion of the generated images.

Complementing the quantitative analysis, qualitative assessment of the output is also conducted to provide a more nuanced understanding of the range and azimuth estimation capabilities of the proposed models.

1.4 Delimitations

The delimitations made in this project include the following:

1. **The two foundation model's parameters are frozen.** The parameters of the two foundation models used to extract semantic and depth information from the RGB images are not fine-tuned on the dataset we use in this thesis. This was done to boost the training speed.

2. **Ignoring Doppler Information.** While radar data provides valuable doppler (velocity) information, our method focuses solely on range and azimuth estimation. We exclude doppler information to maintain a manageable complexity. Processing doppler information from RGB images would require sequence analysis, which would significantly increasing the complexity of the problem.

1.5 List of Publications

This thesis is based on the following publications:

1. **P. A. Rangaraj**, T. Alkanat, and A. Pandharipande, "RAIDS: Radar range- azimuth map estimation from image, depth and semantic

descriptions," in ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (Under Review), 2025.

2. **P. A. Rangaraj**, T. Alkanat, and A. Pandharipande, "RAIDS+: An improved way to generate Radar range- azimuth map estimation from image, depth and semantic descriptions ," IEEE Sensors Journal (Under Review), 2025.

In both these works, I was the lead author and contributor.

Chapter 2

Theory

In this thesis we aim to build a deep learning model that when given an input image, can estimate the range and azimuth position of targets and clutter in the image and use this information to generate a RA map that corresponds to the input image. To provide a comprehensive understanding of the underlying principles, this chapter will first delve into the fundamental concepts of radar range and azimuth estimation. Additionally, we will describe a perceptual evaluation metric and convolutional attention mechanism that is used in this research.

2.1 Radar Range Estimation

Range estimation is fundamental to automotive radars. The range R to a target is determined based on the round trip delay that the Electro Magnetic (EM) waves take to propagate to and from that target.

$$R = \frac{c\tau}{2}, \tag{2.1}$$

where τ is the round-trip time delay in seconds and c is the speed of light in meters per second ($c \approx 3 \times 10^8 \ m/s$). Thus the estimation of τ enables the range measurement.

2.2 Radar Azimuth Estimation

While a radar can estimate range utilising the round-trip time delay, this only provides the information of how far the target is with respect to the radar, it does not provide information about which direction the target is located at. To

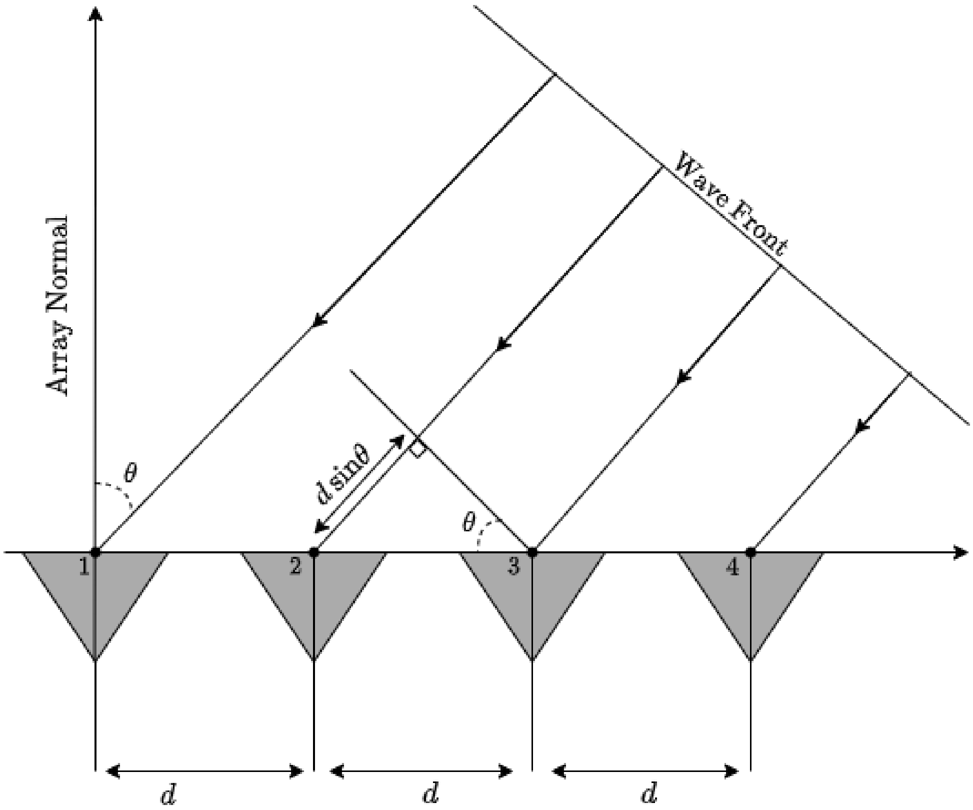


Figure 2.1: Plane wave impinging on a uniform linear array. Adapted from (12)

estimate the angular position of the target, we use the phase shift measurements from multiple receiver antennas. Consider the horizontally oriented uniform linear array shown in Fig.2.1. Each of the gray triangles represent an antenna phase center having its own independent receiver. An EM wave of wavelength λ arrives at the array from an angle of arrival of θ radians relative to the array normal. Consider the wave arriving at receiver no.3 at time t, the wave must travel further to arrive at receiver no.2. This distance can be estimated as,

$$\triangle_d = dsin\theta. \tag{2.2}$$

The time delay $\triangle_\tau$ can then be calculated as,

$$\triangle_\tau = \frac{\triangle_d}{c} = \frac{dsin\theta}{c}. \tag{2.3}$$

The phase shift $\triangle_\phi$ between the two antennas can be calculated using,

$$\triangle_\phi = 2\pi f \triangle_\tau = 2\pi f \frac{dsin\theta}{c} \tag{2.4}$$

where f is the frequency of the EM wave. Modern automotive radars operate at 77GHz, and using $c = f\lambda$ we have,

$$\triangle_\phi = 2\pi f \frac{dsin\theta}{f\lambda} = 2\pi \frac{dsin\theta}{\lambda}. \tag{2.5}$$

From this θ can be estimated as,

$$\theta = sin^{-1}\left[\frac{\lambda}{2\pi}\frac{\triangle_\phi}{d}\right] \tag{2.6}$$

2.3 Structural Similarity Index Metric (SSIM)

Structural Similarity Index Metric (SSIM) (13) is a state-of-the-art perceptual metric, that captures the visual quality of generated images. SSIM is bounded to $\leq$ 1 and is equal to 1 if an only if every pixel matches perfectly. SSIM is based on the philosophy that images are highly structured and their pixels exhibit strong dependencies, and these dependencies carry important information about the structure of the visual scene (13).

The SSIM formula is based on three measurements that are computed for each pixel p in a patch P: luminance (l) eq. (2.8), contrast (c) eq. (2.9), and structure (s) eq. (2.10). The patch P is a local neighbourhood of pixels centered at p. The mean and standard deviation of the ground truth and predicted values are computed over the patch P, using a Gaussian filter G_{σ_G}, with standard deviation σ_G. However, since the Gaussian filter needs to look at neighboring pixels as large as the support of G_{σ_G} the SSIM loss cannot be computed for pixels near the boundaries of the image, where the filter would extend beyond the image boundaries (13). The SSIM score is computed as the product of these three measurements, as shown in eq. (2.7).

$$\text{SSIM}(p) = l(p) \cdot c(p) \cdot s(p). \tag{2.7}$$

The individual comparison functions for each pixel are:

$$l(p) = \frac{2\mu_{Y_P}\mu_{\hat{Y_P}} + C_1}{\mu_{Y_P}^2 + \mu_{\hat{Y_P}}^2 + C_1}, \tag{2.8}$$

$$c(p) = \frac{2\sigma_{Y_P}\sigma_{\hat{Y_P}} + C_2}{\sigma_{Y_P}^2 + \sigma_{\hat{Y_P}}^2 + C_2}, \tag{2.9}$$

$$s(p) = \frac{\sigma_{Y_P\hat{Y_P}} + C_3}{\sigma_{Y_P}\sigma_{\hat{Y_P}} + C_3}, \tag{2.10}$$

where μ_{Y_P} and $\mu_{\hat{Y_P}}$ represent the means of the ground truth and predicted values at patch P, respectively, and $\sigma_{Y_P}^2$ and $\sigma_{\hat{Y_P}}^2$ represent the variances of the ground truth and predicted values at patch P, respectively, with $\sigma_{Y_P\hat{Y_P}}$ denoting the covariance between the two at patch P and C_1, C_2, C_3 are small constants added for stability.

The loss function for SSIM at each patch P is then given by,

$$\mathcal{L}^{\text{SSIM}}(P) = \frac{1}{N}\sum_{p\in P} 1 - \text{SSIM(p)}, \tag{2.11}$$

where N is the number of pixels in patch P. The loss for the entire image I is then given by,

$$\mathcal{L}^{\text{SSIM}}(I) = \frac{1}{N_K}\sum_{P\in K} \mathcal{L}^{\text{SSIM}}(P), \tag{2.12}$$

where K is the set of all patches in image $\mathbf{I}$ and N_K is the number of patches.

2.4 Convolutional Block Attention Module (CBAM)

The importance of attention mechanisms in deep learning models has been extensively studied in previous literature (14; 15; 16; 17). Attention not only guides the model to focus on relevant regions, but also enhances the representation of interests by selectively weighting features (18).

The Convolutional Block Attention Module (CBAM) module consists of two lightweight yet powerful components: the channel attention module and the spatial attention module. Given an intermediate feature map $F \in \mathbb{R}^{C\times H\times W}$, CBAM infers a 1D channel attention map $M_c \in \mathbb{R}^{C\times 1\times 1}$ and a 2D spatial attention map $M_s \in \mathbb{R}^{1\times H\times W}$, which are then processed using a simple multi-layer perceptron (MLP) and a convolution layer, respectively.

The resulting attention maps are then element-wise multiplied with the feature map, effectively weighting the features based on their importance. Notably, the authors of (18) found that performing channel attention first and then spatial attention yields slightly better results than performing spatial attention first, highlighting the importance of careful attention mechanism design.

The overall attention process can be summarized as:

$$F' = M_c(F) \otimes F, \tag{2.13}$$

$$F'' = M_s(F') \otimes F', \tag{2.14}$$

where $\otimes$ denotes element-wise multiplication.

Chapter 3

Background

This chapter provides a comprehensive review of the existing literature on synthetic radar data generation, with a focus on three key works that are relevant to our proposed method. We begin by examining the use of Generative Adversarial Networks (GANs) to generate Range-Doppler (RD) maps from noise inputs, highlighting the potential and drawback of this approach for generating synthetic radar data. We then discuss cross-modality conversion methods, including the generation of Range-Azimuth (RA) point clouds from LiDAR point clouds and another method from RGB images, which demonstrate the feasibility of converting data from one sensor modality to another. By reviewing these related works, we establish a foundation for understanding the current state-of-the-art in synthetic radar data generation and the key challenges and opportunities in this field.

3.1 Range Doppler maps using generative adversarial networks

Recent advances in GANs have enabled the synthesis of realistic images (19; 20). In the context of radar data generation, GANs have been employed to create synthetic RD maps of Frequency-Modulated Continuous-Wave (FMCW) radar for short-range scenarios involving pedestrians and cyclists (21). Similarly, Conditional Generative Adversarial Networks (CGANs) have been utilized to synthesize RD maps with Micro-Doppler signatures for a selected object input (22). However, these approaches are limited by their scalability and versatility, primarily due to their reliance on manual object selection and the need for extensive retraining for each new object or scenario. This makes these approaches ill suited for generating large amounts of synthetic radar data.

This limitation underscores the need for a more robust and scalable approach for generating synthetic radar data, which can adapt to diverse scenarios and objects without requiring extensive retraining.

The primary focus of this thesis is to address this limitation by developing a novel approach for generating synthetic radar data that can adapt to diverse scenarios and objects without requiring any retraining.

3.2 LiDAR point cloud to RA point cloud

LiDAR point clouds possess a rich source of depth information, albeit with limited semantic information due to their inherent sparsity. The acquisition of dense depth maps from LiDAR point clouds is often hindered by the requirement for high-resolution LiDAR systems, which can be prohibitively expensive. Nevertheless, the depth information extracted from sparse LiDAR point clouds can still be leveraged to construct a RA point cloud, a critical component of radar perception systems. A RA map and RA point cloud are two distinct representations of radar data. While both formats contain information about the range and azimuth of radar reflections, they differ in their sparsity: RA point clouds are typically processed using detection algorithms that remove clutter and preserve only areas of high reflectivity, or ”hot spots,” in the scene which results in a sparse view. In contrast, RA maps retain all radar reflections, providing a more comprehensive view of the environment.

Recent research has explored the feasibility of translating LiDAR data to radar RA point cloud using GANs (8). This approach employs a combination of local and global generators, along with region of interest extractors, to facilitate the translation process. The local generator utilizes a U-Net framework (23), while the global generator is guided by an occupancy grid and the encoder from PointPillars (24) is employed to extract regions of interest from LiDAR point clouds.

The existing approach of translating sparse LiDAR point clouds to radar RA point clouds using GAN (8) is inherently limited by the sparsity of the input data, resulting in sparse radar point clouds. In contrast, our method capitalizes on the dense nature of RGB images to produce dense RA maps, providing a richer and more informative representation of the scene for downstream tasks. This fundamental difference in input representation and output density enables our approach to offer a more comprehensive understanding of the environment, thereby enhancing the performance of subsequent tasks.

3.3 RGB images to RA point cloud

A recent study (4) proposed a convolutional neural network (CNN) based method for emulating radar point cloud data from RGB images, effectively establishing a pipeline for converting images to sparse point clouds. This approach leverages a ResNet (25) CNN backbone to estimate the parameters of a Gaussian mixture model, which specify a radar point cloud probability density function (PDF) corresponding to the input camera image. The resulting PDF model represents the likelihood of presence of a radar point on arbitrary spatial coordinates. This information can then be used to generate multiple realizations of a radar point cloud corresponding to the input camera image by sampling the PDF.

Notably, the approach proposed in (4) represents the current state-of-the-art in using RGB images as input to generate synthetic radar data. However, our method differs from this approach in a key aspect: as the training of the model in (4) is guided by sparse radar point clouds, their model also generates sparse point clouds, our method aims to produce dense representations of RA maps from RGB images, thereby providing a more comprehensive and detailed understanding of the scene.

Chapter 4

Methodology

In this chapter, we introduce Range Azimuth from Image, Depth and Semantic descriptors (RAIDS), a novel method that leverages camera RGB images, dense depth maps and semantic information to synthesize high-quality dense RA maps. RAIDS uses a Convolutional Auto Encoder (CAE) architecture that is inspired by the well known U-Net network (23). We motivate our architectural choices and also talk about the key additions that were used to improve the quality of the generated RA maps. We will also focus on the loss function that places priority on both the pixel wise accuracy and the perceptual quality of the generated RA map.

4.1 Using Depth maps and Semantic Segmentation

In this work, we propose the incorporation of depth maps and semantic segmentation maps as supplementary sources to the input RGB image. Although these maps are derived from the same RGB source as the input images, they provide valuable contextual information that can be leveraged to enhance the accuracy of our model. From an information-theory perspective, it may seem counter-intuitive to use these supplementary sources, as they do not introduce new information. However, a growing body of research has consistently demonstrated that the use of depth and semantic information can significantly enhance the performance of computer vision tasks.

For instance, studies have shown that incorporating depth information into semantic segmentation algorithms can yield better segmentation performance (26). Conversely, utilizing semantic information has been found to improve the accuracy of monocular depth estimation models (27; 28). These findings suggest that the fusion of depth and semantic information can provide a synergistic effect, enabling the model to better understand the scene and make more accurate predictions. This synergy is rooted in the fact that depth information provides a 3D representation of the scene, while semantic information provides a higher-level understanding of the scene's contents.

By leveraging these supplementary sources, our model can tap into this rich contextual information, ultimately leading to improved performance and more accurate predictions. The use of depth and semantic maps as supplementary sources is a deliberate design choice, motivated by the desire to provide our model with a more comprehensive understanding of the scene. Furthermore, by using pre-trained foundation models that have been trained on massive amounts of data, we enable our model to leverage the collective knowledge and patterns learned from these large-scale datasets, thereby unlocking the full potential of our model, and achieving state-of-the-art performance in radar data generation.

The objective of RAIDS is to generate RA maps from RGB images. To achieve this, the model must be able to infer range or depth information from the image, estimate the azimuth of the targets in the scene, and predict the strength of the radar reflection for each target.

To accurately predict the strength of the radar reflection, we must consider that radar reflections are material-dependent. We utilise semantic maps to capture this dependency. By incorporating the semantic map, we aim to capture class-dependent variations in radar reflectivity, thereby enhancing the accuracy of the generated RA maps.

By leveraging depth maps as an additional input, the model can use this to infer range information. Additionally, azimuth information can be extracted from RGB images, which provide a dense representation of the scene's azimuthal structure. The combination of depth and semantic information with RGB images enables the model to effectively estimate the range, reflection strength and azimuth of targets in the scene, ultimately generating accurate RA maps.

4.2 Architecture of RAIDS

4.2.1 Depth Map

Accurate depth map generation is a critical component of our proposed model for producing high-quality RA maps from RGB images. The generated depth map must effectively recover the three-dimensional geometry of a scene from a single two-dimensional image, thereby facilitating the computation of distances between the camera and objects within the scene. In this study, we investigate two distinct approaches for generating depth maps: relative depth estimation or Monocular Depth Estimation (MDE) and absolute depth estimation or Monocular Metric Depth Estimation (MMDE).

Relative depth estimation algorithms output a single-channel image where each pixel represents a relative distance from the camera to a point in space, without referencing real-world units of measurement, commonly known as disparity value. We utilize the Depth Anything model (29) to obtain relative depth maps, which leverages semantic aided perception to enhance the accuracy of the estimated depth. This technique integrates high-level semantic information into the depth estimation process, allowing for more accurate and robust depth estimates. It is important to note that the generated map from Depth Anything (29) contains disparity values that are inversely proportional to the true depth by an unknown scaling and shift factor.

In contrast, absolute depth estimation algorithms output a single-channel image where each pixel represents an absolute distance from the camera to a point in space, measured in real-world units such as meters. We use the UniDepth model (10) to obtain absolute depth maps, thereby decoupling the errors from the respective branches. The UniDepth model utilizes cross-attention layers to share information between the branches and can utilize the camera's intrinsic parameters to generate accurate and reliable depth maps.

Both relative and absolute depth maps can provide valuable information for disentangling targets in the RGB image. However, when working with absolute depth maps the model can leverage the direct correspondence between the depth values and the RA domain, enabling more accurate and reliable depth estimates. In contrast, relative depth estimation methods require the model to perform an additional inference step of estimating the scaling and shifting factor, which can introduce errors and reduce the overall accuracy of the depth estimates.

Our proposed model, RAIDS, uses UniDepth (10) to generate dense metric depth maps, leveraging the direct correspondence between the depth values and the RA domain to enable accurate and reliable depth estimates. This design choice allows RAIDS to focus on transferring RGB images to the RA domain, without the added complexity of inferring scale and shift factors which vary for each image, and ultimately produces high-quality RA maps.

4.2.2 Semantic segmentation

To accurately estimate the strength of radar reflections, we require contextual information about the scene. Semantic segmentation maps, which categorize each pixel in an image into a class or object, provide a rich source of information about the scene's composition and layout.

To generate semantic maps for each RGB input, we utilize the pre-trained Mask2Former foundation model (11), which is built upon the Swin transformer backbone (30). The Mask2Former model has been fine-tuned on the CityScapes dataset (6), a comprehensive dataset that focuses on the semantic understanding of urban street scenes. The Mask2Former model can segment each scene into 19 classes, corresponding to seven groups of the CityScapes dataset. By incorporating the resulting semantic map into our model, we can leverage the rich semantic information it provides to inform our estimates of radar reflection strength and improve the overall accuracy of our generated RA maps.

Notably, our selection of Mask2Former was informed by its utilization of multi-scale high-resolution features, which enables the detection of distant targets that appear small in the scene (11). This is particularly relevant in the context of RA map generation, where the ability to detect and accurately represent distant targets is crucial. As demonstrated in Fig 6.2 Scene 5, the Mask2Former model is able to effectively detect and segment distant targets, thereby proving to be a robust source of semantic information for our RA map generation pipeline.

4.2.3 Convolutional Auto Encoder

To generate an RA map, we employ a Convolutional Auto Encoder (CAE) architecture inspired by the well-known U-Net network (23), as illustrated in Fig.4.1. This design choice is motivated by the U-Net's strong localization capabilities, which arise from the combination of low-level features from convolutions and high-level features from residual connections. This feature is particularly desirable for our task, as we aim to localize object activations to the correct location in the range and azimuth dimensions.

Our Convolutional Auto Encoder (CAE) architecture features a three-stage encoder and decoder with a bottleneck stage, where each stage is equipped with a CBAM attention block to enhance the localization capabilities of the CAE and improve the detection of targets from the input features. The

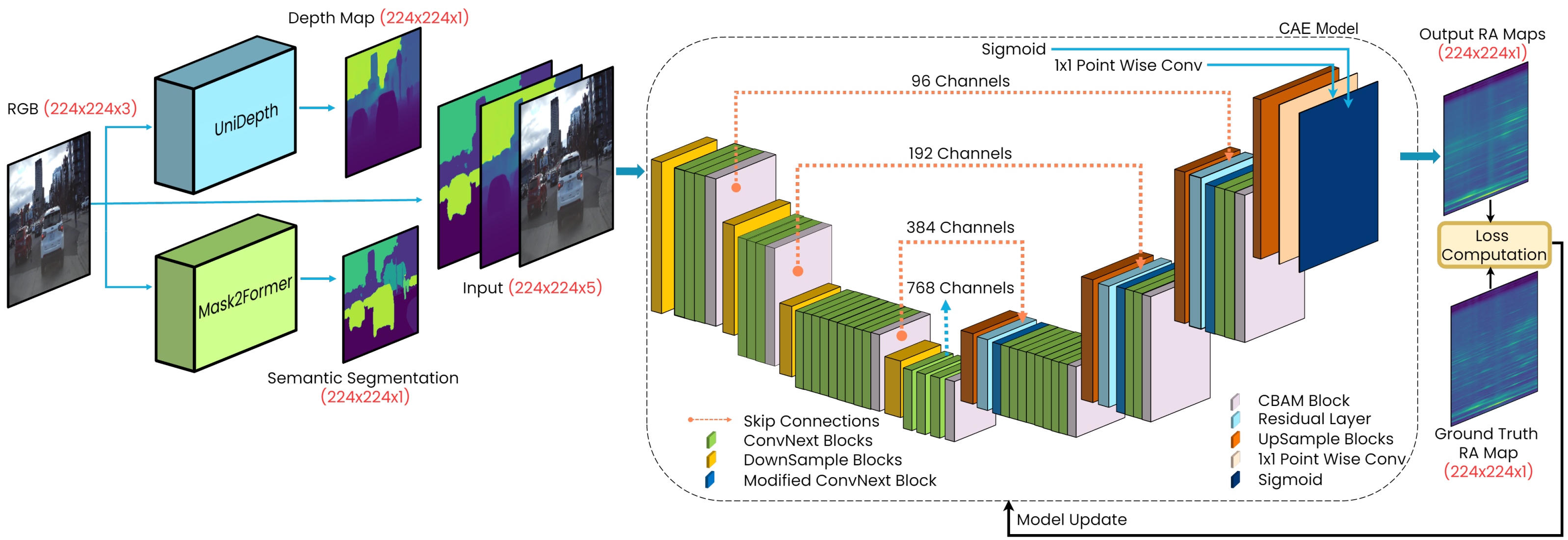


Figure 4.1: The proposed RAIDS model uses camera images, as well as monocular metric depth map and semantic map obtained from UniDepth and Mask2Former foundation models respectively, to generate RA maps through a hierarchical encoder-decoder architecture. ConvNext blocks facilitate efficient feature extraction and compression in the encoder and RA map reconstruction in the decoder, while preserving crucial features and high-frequency details. CBAM offers spatial and channel attention that prioritises and localises targets better.

CBAM module helps to selectively focus on the most relevant features and suppress the less important ones, resulting in a more accurate and robust feature representation.

The encoder and decoder are connected by residual connections, which enable the decoder to access information from the corresponding encoder layer. This allows the decoder to retain key features from the encoder and prevents them from being lost in the bottleneck layer. Both the encoder and decoder are built using ConvNext blocks (31), which provide a robust foundation for feature extraction and representation. Notably, the ConvNext blocks in our architecture operate at a higher receptive field with a kernel size of 7×7, allowing for a more comprehensive understanding of the input features. To accommodate the residual feature, the first ConvNext block of each expanding module in the decoder is modified to process twice the number of channels, resulting in the "Modified ConvNext block" shown in Fig. 4.1.

Finally, a sigmoid transformation is applied to the output of the decoder to ensure that the final estimate of pixel value lies within the range of 0-1.

4.3 Loss Function

The goal of RAIDS is to synthesize pixel-wise accurate and visually similar RA maps that correctly estimate the range azimuth of targets. For this reason, we believe our loss function should be motivated by a combination of pixel-wise and perceptual metric. We use a loss function that was introduced in (32), it is a combination of a pixel wise metric namely ℓ_1 loss, and a perceptual metric namely the Multi-Scale Structural Similarity Index Metric (MS-SSIM). We chose ℓ_1 loss over MSE because it provides a better trade-off between smoothness and detail preservation, allowing the model to retain the important high-frequency regions in the generated RA maps.

The formula for ℓ_1 loss is given by,

$$\mathcal{L}^{\ell_1} = \frac{1}{n} \sum_{i=1}^{n} |Y_i - \hat{Y}_i|, \tag{4.1}$$

where n is the total number of pixels, Y_i is pixel value of the ground truth at position i and $\hat{Y}_i$ is pixel value of the predicted image at position i.

The MS-SSIM is based on the SSIM loss described in section 2.3. From eq. (2.8), eq. (2.9) and eq. (2.10) we know that the computation of SSIM(p)

requires looking at a neighbourhood of pixel p as large as the support of the Gaussian filter G_{σ_G}. This means that the choice of σ_G which is the standard deviation of the Gaussian filter has an impact on the quality of the processed result (32). It was found in (32), that for smaller values of σ_G the ability to preserve local structure is lost and for higher values of σ_G the high frequency regions are preserved. Instead of optimizing for the best σ_G, (32) proposes computing the SSIM(p) for each pixel p with M different values for σ_G. In each of the M levels, the values for σ_G is half of the previous. This loss is then called Multi-Scale Structural Similarity Index Metric (MS-SSIM). The formula for MS-SSIM is then given by,

$$\mathcal{L}^{\text{MS-SSIM}}(p) = 1 - \left(l_M(p) \cdot \prod_{j=1}^{M} c_j s_j \left(p\right) \right), \tag{4.2}$$

where luminance (l_M) is computed at scale M using eq. (2.8) and contrast (c_j) and structure (s_j) are computed at scale j, where $j = \{1, ..., M\}$ using eq. (2.9) and eq. (2.10).

As explained in section 2.3, the SSIM loss cannot be computed for pixels near the boundaries of the image as the Gaussian filter needs to look at neighboring pixels as large as the support of G_{σ_G}, which may extend beyond the image boundaries. Given the convolutional nature of our model, the learned kernel is applied to all pixels in the image, and the gradients of the loss function flow back to all pixels within the support of G_{σ_G}, including to those near the boundaries. As a result, even though the MS-SSIM loss is not computed for boundary pixels, the gradients still flow back to these pixels, allowing them to be updated during training. This is a crucial aspect of our loss function, as it ensures that all pixels in the image, including those near the boundaries, are updated and refined during the training process.

Finally, the combined loss function is a weighted sum of the MS-SSIM and ℓ_1 loss:

$$\mathcal{L}^{Mix} = \alpha \mathcal{L}^{\text{MS-SSIM}} + (1 - \alpha)\mathcal{L}^{\ell_1} \tag{4.3}$$

4.4 Baseline Model

We create a baseline model to serve as a direct comparison to our proposed RAIDS model. There currently exists no other method that generates RA maps from RGB images in a similar manner. This baseline model uses only the RGB image as input, without incorporating any additional features or attention mechanisms. Specifically, the baseline model does not utilize the CBAM, which is a key component of the proposed RAIDS model. By comparing the performance of the baseline model to the proposed RAIDS model, we can both quantitatively and qualitatively assess the impact of incorporating multiple input strategies and spatial and channel attention mechanisms on the generation of accurate RA maps. The quantitative evaluation will provide a numerical measure of the performance improvement, while the qualitative evaluation will allow us to visually inspect and compare the generated RA maps, providing insight into the differences in their quality and accuracy. The architecture of the baseline model is illustrated in fig. 4.2.

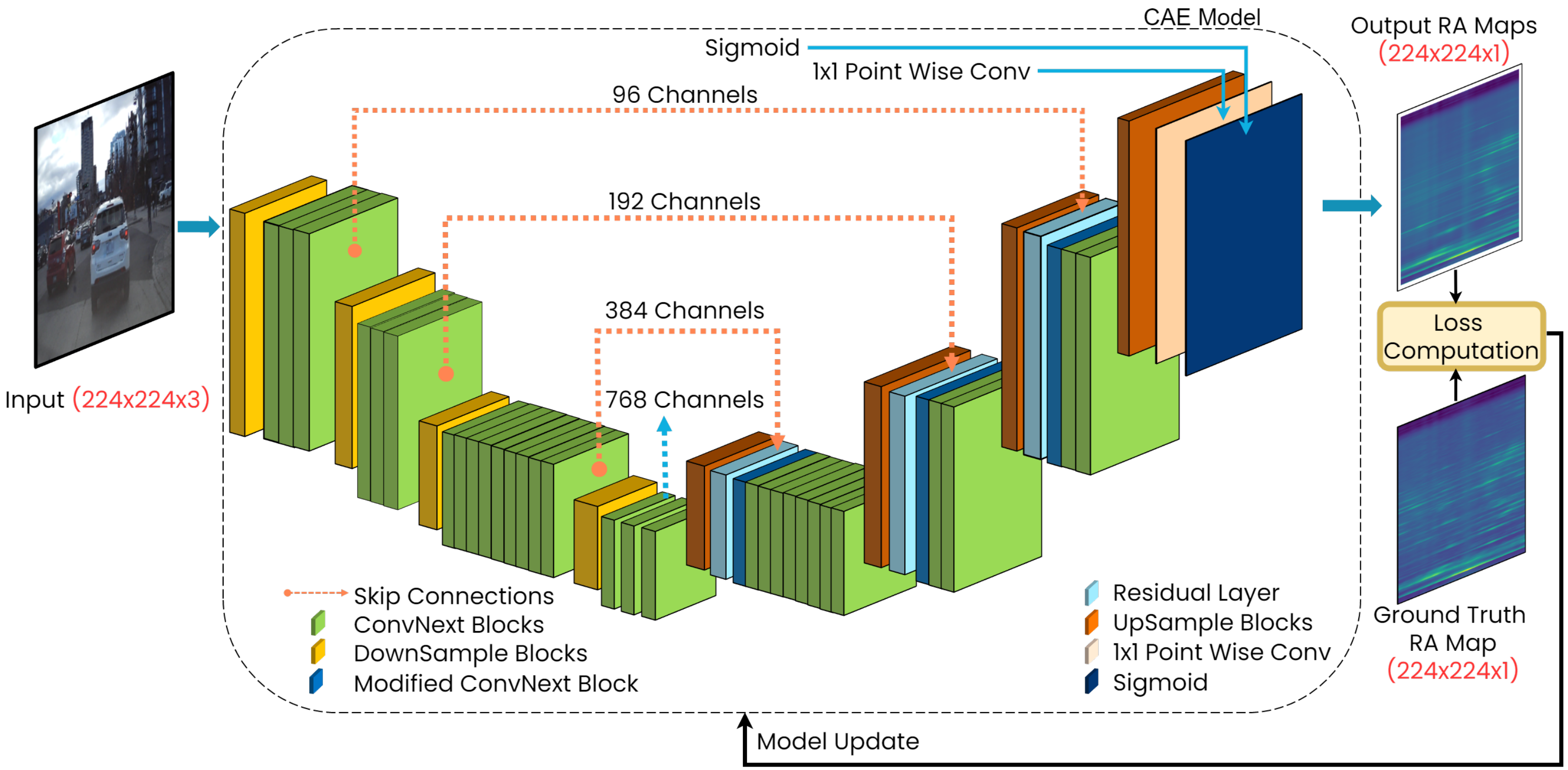


Figure 4.2: The baseline model only uses the RGB images to generate RA maps through a hierarchical encoder-decoder architecture. ConvNext blocks facilitate efficient feature extraction and compression in the encoder and RA map reconstruction. CBAM is not used in the baseline model.

Chapter 5

Dataset and Evaluation

The proposed models were trained and evaluated utilizing the publicly available RADIaL dataset (33). To further evaluate the generalizability of our models, we also conduct qualitative tests on the CRUW dataset (34) through a transfer learning approach, where the learned weights are directly applied to the CRUW dataset without fine-tuning. This allows us to assess the generalizability of our models in a more challenging scenario, as the CRUW dataset features a different radar configuration, diverse scenarios, and different maximum range.

We conduct an ablation study to examine the contribution of individual components to our model's performance, followed by a comprehensive analysis of the resulting RA maps, comprising both qualitative and quantitative evaluations. Our quantitative evaluation metrics include MSE, ℓ_1 loss, Peak Signal to Noise Ratio (PSNR), and Structural Similarity Index Metric (SSIM), providing a thorough understanding of the models' performance.

5.1 Model Setup

We use PyTorch (35) and AdamW (36) to build and optimize our model. To adapt the pre-trained ConvNext model to a five-channel input, we duplicate the first layer of the model and modify it to accept five channels. The first three channels are assigned the weights from the pre-trained ConvNext model (31) for processing the RGB images. Weights of the remaining two channels are initialized using the Xavier initialization strategy (37). The initial learning rate is set to 4×10^{-4} for the encoder and 4×10^{-3} for the decoder. We experiment with different values for the weighting factor α in our loss function eq. (4.3), but do not perform a thorough investigation. However, we observe that the results are not significantly sensitive to small variations of α. Based on this observation, we set α to 0.84 in our experiments. For the computation of the MS-SSIM loss in eq. (4.2), we use a set of Gaussian filters with standard deviations $\sigma_G = \{0.5, 1, 2, 4, 8\}$, as recommended in (32). We use cosine annealing (38) to reduce the learning rate and perform warm restarts and DropPath (39) is used to regularise the model, which helps to improve convergence. The model is trained for 100 epochs.

To prepare the ground truth RA map data for training, we need to normalize and modify it to ensure stable and meaningful learning. The raw RA map data from the RADIaL dataset (33) exhibits a highly skewed distribution, with the vast majority of values concentrated in a very narrow range. Specifically, the data is characterized by a long tail of extremely large values, which poses a challenge for normalization. When we attempt to normalize this data using min-max normalization, we find that the resulting distribution is highly uneven, with most values clustered near zero. To address this issue, we first apply a log transformation to the RA map data. This transformation has a profound impact on the data distribution, compressing the extreme values and revealing a more nuanced structure. After applying the log transformation, we min-max normalize the RA map data to ensure that it lies within the range of 0-1.

5.2 RADIaL Dataset

We utilize the publicly available RADIaL dataset (33) for training our model. This dataset comprises of 91 sequences, and a total of 25,433 RGB image - RA map pairs, with a maximum temporal separation of 20ms, and are positionally aligned. The RGB image was resized to a resolution of 224×224 for efficient computation. We also tested out 512×512, but did not notice any improvement in using a higher input resolution. The dataset encompasses diverse environments, including highways, urban streets, and rural roads, thereby ensuring a comprehensive training and testing environment. The RADIaL dataset was collected using an imaging radar with a large MIMO (multiple-input multiple-output) array of 12×16 antennas, featuring a maximum range of 103m. Upon examining the RADIaL dataset, we observed that the rows corresponding to ranges between 93 and 103 meters were corrupted in the ground truth, prompting us to crop the data to a maximum range of 91m. Additionally, we filtered out radar points with azimuth values outside the interval [-50°, +50°] to align the radar and camera fields-of-view. The dataset was then split into training and test sets. To ensure that the test set is diverse and representative of the various environments and scenarios present in the dataset, we selected sequences with less than 100 frames to be included in the test set, resulting in 24 diverse sequences with 1,426 images upon which the models are evaluated. The remaining 67 sequences with 24,007 images were assigned to the training set. We normalized the RGB images using the ImageNet (40) mean and deviation to ensure consistency across the dataset. Furthermore, we normalized the generated depth and semantic segmentation maps to be in the range of 0-1. Specifically, we identified the maximum and minimum estimated depth across the entire dataset to normalize the depth map. The semantic segmentation map, which contains integer values between 0 and 18 representing the 19 classes, was min-max normalized such that the lowest class (0) corresponds to 0 and the highest class (18) corresponds to 1 in the normalized map.

5.3 CRUW Dataset

The CRUW dataset (34) is a publicly available camera-radar dataset for autonomous vehicle applications, which differs from the RADIal dataset in several key aspects. Most importantly, CRUW uses a different radar system, which has a maximum range of 25m and a field of view of [-90°, +90°] (34). In contrast the RADIaL dataset has a maximum range of 103m and field of view of [-50°, +50°]. Furthermore, CRUW covers a range of scenarios, including parking lots, campus roads, city streets, and highways, which is more diverse than the RADIaL dataset that mostly consists of urban, rural, and highway scenes with automotive vehicles and has very few scenes with pedestrians or cyclists. To evaluate the generalizability of our models, we randomly sampled 4,000 scenes from the CRUW dataset which contains a total of 40,000 diverse scenes (34). This subset of scenes was then used as a test set to evaluate the performance of our proposed model, providing a comprehensive assessment of their ability to generalize to unseen environments.

5.4 Evaluation Setup

To evaluate the performance of our proposed model, we conduct an ablation study to investigate the added benefit of each supplementary information source from different foundation models. This involves systematically testing different combinations of input strategies and evaluating the performance of various depth estimation models. Once the optimal model is identified, we use it to conduct a thorough evaluation of its performance. We perform both qualitative and quantitative analyses on the RADIaL dataset and qualitative analysis on the CRUW dataset. This evaluation setup allows us to assess the impact of supplementing the RGB image with depth and semantic information, and to gain insight into the importance of these auxiliary inputs.

5.4.1 Ablation Study

To comprehensively investigate the impact of each scene descriptor on the model output, we conduct a thorough ablation study by testing different input data strategies. Specifically, we compare the performance of our proposed RAIDS model, which utilizes a combination of RGB images, semantic maps, and metric depth maps as input, with a baseline model that only relies on RGB images. Additionally, we introduce four variants of our proposed model that use different combinations of scene descriptors as input and do not use CBAM:

1. RGB images + semantic maps from Mask2Former (11)
2. RGB images + relative depth maps from Depth Anything (29)
3. RGB images + metric depth maps from UniDepth (10)
4. RGB images + semantic maps + relative depth maps, which we refer to as the "intermediate model"

This allows us to not only assess the contribution of each individual scene descriptor, but also to evaluate the importance of using metric depth versus relative depth in our model, as well as the impact of combining different scene descriptors. We evaluate the performance of these variants using two widely adopted metrics: PSNR and SSIM. The evaluation is performed on the RADIaL test set (33), and the average results for each variant are presented, providing a comprehensive understanding of the contribution of each scene descriptor to the model's output and the impact of using metric versus relative depth on the overall performance.

5.4.2 Quantitative Analysis

Pixel based error computation is a very common evaluation metric in image processing tasks. In addition to the proposed model and the baseline model, we also compare the generated RA maps from the intermediate model which was introduced in our ablation study, on two pixel based metrics, the MSE which is defined below and ℓ_1 loss which is defined in eq. (4.1).

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2, \tag{5.1}$$

where n is the total number of data points, Y is ground truth value and $\hat{Y}$ is the predicted value.

The spatial structure and pixel intensity values inherent to RA maps render them amenable to evaluation using distortion assessment metrics such as Peak Signal to Noise Ratio (PSNR), commonly employed in signal processing applications. PSNR represents the ratio between the maximum possible value (power) of a signal and the power of distorting noise that affects the quality of its representation and is given by,

$$\text{PSNR} = 20 \cdot \log_{10}(\text{MAX}_I) - 10 \cdot \log_{10}(\text{MSE}), \tag{5.2}$$

where MAX_I is the maximum pixel values of the original image. The calculated PSNR values is output in decibels (dB).

In addition to the objective evaluation using PSNR, we also use Structural Similarity Index Metric (SSIM) defined in eq. (2.7) as a perceptual evaluation metric to assess the fidelity of the synthesized RA map of the proposed and baseline model.

5.4.3 Qualitative Analysis

To conduct a qualitative evaluation of our models, we carefully select a diverse set of scenes from the RADIaL (33) and CRUW (34) datasets. The RADIaL dataset provides a wide range of scenes, including highways, urban streets, and rural roads, each with varying levels of complexity and clutter. We choose scenes that exhibit different target densities and types of clutter, such as vegetation and buildings, to comprehensively assess the performance of our models.

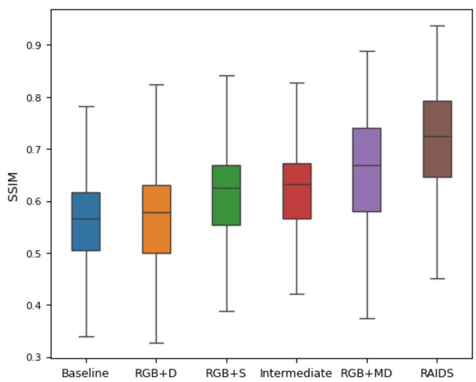
In contrast, the CRUW dataset offers a unique opportunity to evaluate our models on scenes that contain target classes that are underrepresented in the RADIaL dataset, which was used to train our baseline and proposed models. Specifically, we select scenes that feature pedestrians and cyclists in parking lots, as well as a highway scene that is comparable to the data from the RADIaL dataset. This allows us to assess the generalizability of our models to new and unseen scenarios.

Through qualitative analysis, we gain valuable insights into the range and estimation capabilities of each model. By examining the nuances of each model's performance, we can identify their strengths and weaknesses, and develop a deeper understanding of their behavior in different scenarios. This qualitative evaluation provides a complementary perspective to our quantitative results, and offers a more comprehensive understanding of the performance of our models.

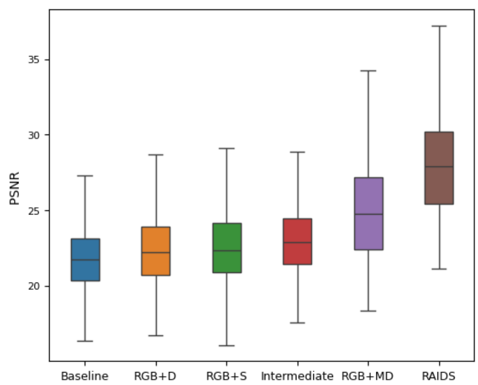
Chapter 6

Results and Analysis

6.1 Ablation Study



(a) SSIM



(b) PSNR

Figure 6.1: Ablation results with different input data strategies; RGB, D, MD and S represent camera image, depth map from Depth Anything, metric depth map from UniDepth and semantic segmentation map from Mask2Former, respectively.

We present the results of our ablation study in Fig.6.1a and Fig.6.1b, where we compare the average SSIM and PSNR of six different model variants across the entire test set. Our analysis reveals that the RAIDS model achieves the highest mean in both metrics, outperforming all other variants and demonstrating its superior performance in generating accurate RA maps. Conversely, the baseline model, which only uses RGB images as input, achieves the lowest mean in both metrics, highlighting the limitations of relying solely on RGB

	Metric	Scene 1	Scene 2	Scene 3	Scene 4	Scene 5	Overall
Baseline	L1	0.048	0.043	0.051	0.046	0.036	0.058
	SSIM	0.561	0.590	0.475	0.486	0.595	0.563
	PSNR	23.9	24.2	22.6	22.8	25.1	21.1
	L2	0.004	0.005	0.005	0.005	0.003	0.007
Intermediate	L1	0.042	0.040	0.039	0.053	0.029	0.049
	SSIM	0.572	0.617	0.533	0.515	0.616	0.622
	PSNR	24.3	24.4	23.1	22.3	26.6	22.9
	L2	0.003	0.005	0.004	0.005	0.002	0.005
RAIDS	L1	**0.026**	**0.017**	**0.031**	**0.003**	**0.020**	**0.026**
	SSIM	**0.591**	**0.630**	**0.542**	**0.542**	**0.644**	**0.717**
	PSNR	**28.8**	**31.2**	**26.1**	**26.03**	**29.23**	**28.2**
	L2	**0.001**	**0.0007**	**0.002**	**0.002**	**0.001**	**0.001**

Table 6.1: Quantitative results of the baseline, intermediate and our proposed model on the different scenes shown in Fig.6.2.

data for RA map generation.

Notably, our results show that the variants with partial scene descriptors as input exhibit incremental improvements in performance, with the addition of relative depth and semantic segmentation leading to moderate gains in SSIM and PSNR. However, the most significant finding is observed when using metric depth: the model that utilizes RGB + metric depth achieves a monumental performance gain, outperforming even the intermediate model that combines RGB, semantic, and relative depth. This suggests that the utilization of metric depth is a crucial factor in achieving high-quality RA maps, and that its impact is more pronounced than the combination of other scene descriptors. The results of our ablation study provide strong evidence for the effectiveness of our proposed approach and demonstrate the value of incorporating scene descriptors, particularly metric depth, into the RA map generation process.

6.2 Performance evaluation on RADIAL

In Fig.6.2, we present the qualitative results of the proposed, intermediate and baseline models for five selected scenes. These scenes are used to illustrate the performance of the models in different scenarios. In table 6.1, we provide

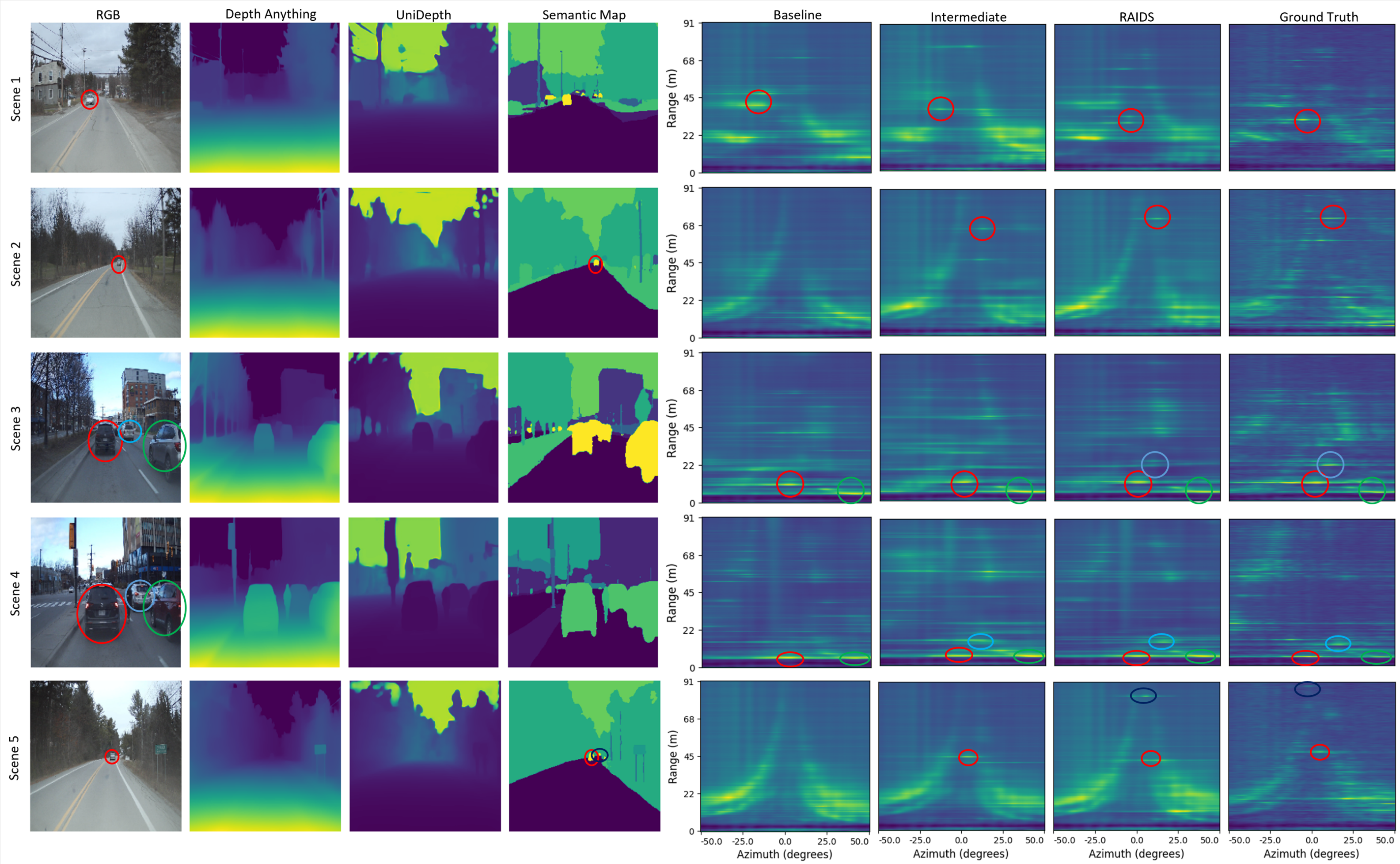


Figure 6.2: Qualitative result on the RADIaL dataset. From left to right, each column shows the camera images, estimated depth map from Depth Anything, estimate depth map from UniDepth, semantic segmentation map from Mask2Former, estimated RA maps from the baseline model, RA maps generated by the intermediate model, RA maps generated by CAE, and the ground truth RA maps for three different scenes. Every color represents a different object class in the semantic segmentation maps. Key detections are circled in the RA map and small targets are circled in the RGB image. Scenes with multiple targets have colour coded circles. Figure best viewed in color.

the quantitative results for these models on the selected scenes and the entire test set.

In Fig.6.2 Scene 1, we observe that RAIDS accurately detects the oncoming car and estimates its range to be 38m (activation corresponding to red circle), which is in close agreement with the ground truth. In contrast, the intermediate model overestimates the range to be 45m, demonstrating a discrepancy in range estimation accuracy. This discrepancy can be attributed to the utilization of a metric depth estimation model in RAIDS, which provides a more accurate representation of the scene and enables improved range estimation capabilities.

In Fig.6.2 Scene 2, we observe that the RAIDS model accurately detects the oncoming car (activation corresponding to red circle) and estimates its range to be in close agreement with the ground truth. In contrast, the intermediate model underestimates the range of the car, highlighting the limitations of its range estimation capabilities. Furthermore, the baseline model struggles to detect the target car, which is located at a distance. This underscores the challenges faced by the baseline model in handling targets that are far away.

The results for Fig.6.2 Scene 3 provide compelling evidence of RAIDSs ability to detect and localize multiple targets in close proximity. Notably, while all three models detect the two nearby cars (activations corresponding to red and green circles), only RAIDS successfully detects the third car (activation corresponding to the blue circle) and accurately estimates its range, achieving a high degree of agreement with the ground truth. This performance can be attributed to the strategic incorporation of CBAM attention blocks in RAIDS, which enables the model to selectively focus on specific areas of the image and semantic classes, thereby enhancing its detection capabilities.

Furthermore, the results for Fig.6.2 Scene 4, demonstrate RAIDS ability to accurately detect and localize multiple targets in a complex scene. Specifically, RAIDS exhibits strong activations for the three cars in close proximity (activations corresponding to red, green and blue circles), as well as a weak activation near the 68m mark for the clutter, which is consistent with the ground truth. In contrast, the intermediate model is able to detect the cars, but produces a much stronger activation in the 68m range, indicating that it wrongly estimates the strength of clutter activations. The superior performance of RAIDS in this regard can be attributed to its enhanced ability to relate the strength of the activation to the distance of the target, a capability that is facilitated by the use of CBAM attention blocks.

In Fig.6.2 Scene 5, both the intermediate model and RAIDS accurately detect the car (activation corresponding to red circle) at the 50m mark, with range estimations that are very close to the ground truth. However, the intermediate model fails to detect a distant car that is barely visible in the RGB image but is more apparent in the semantic map (activation corresponding to the black circle in semantic map). In contrast, the RAIDS model successfully detects this car and places its target activation near the 91m mark, consistent with the ground truth. This demonstrates the improved performance of RAIDS in detecting distant targets, which can be attributed to the use of CBAM attention blocks and the metric depth map from UniDepth (10).

These improvements are also consistent with the qualitative results, as RAIDS achieves the highest value for the five scenes, as shown in Table.6.1. Furthermore, this behaviour extends across the entire test set of 1,426 test images, as evident from the quantitative results in Table.6.1, demonstrating the robustness and reliability of RAIDS in detecting and localizing targets in various scenes. Finally, in Fig. 6.1, we present The quantitative results which confirm the superiority of RAIDS, which achieves the highest mean in both SSIM (Fig.6.1a) and PSNR (Fig.6.1b) metrics.

6.3 Performance evaluation on CRUW

Our evaluation on the CRUW dataset (34) provides a rigorous assessment of the robustness and generalizability of our models in diverse scenarios that differ from the RADIal dataset. Notably, our models exploit the knowledge learned from the RADIal dataset, without any fine-tuning on the CRUW dataset, thereby providing a true test of their ability to generalize to novel, unseen environments.

In Fig.6.3 Scene 1, we observe that the baseline model, which relies solely on RGB images, fails to detect pedestrians, a low-probability class in the RADIal dataset. In contrast, the intermediate model and RAIDS models successfully detect pedestrians (activations corresponding to red and green circles), leveraging the availability of semantic information. Furthermore, the RAIDS model achieves a more accurate range estimate than the intermediate model, thanks to the incorporation of a metric depth map.

In Fig.6.3 Scene 2, we evaluate the performance of our models on a scene containing a cyclist, another low-probability class. While all three models exhibit some activation (corresponding to the red circle) near the 6.25m mark, the baseline and intermediate models produce non-localized activations that extend into nearby azimuth bins or exhibit multiple activations at different ranges. In contrast, the RAIDS model produces a localized target activation

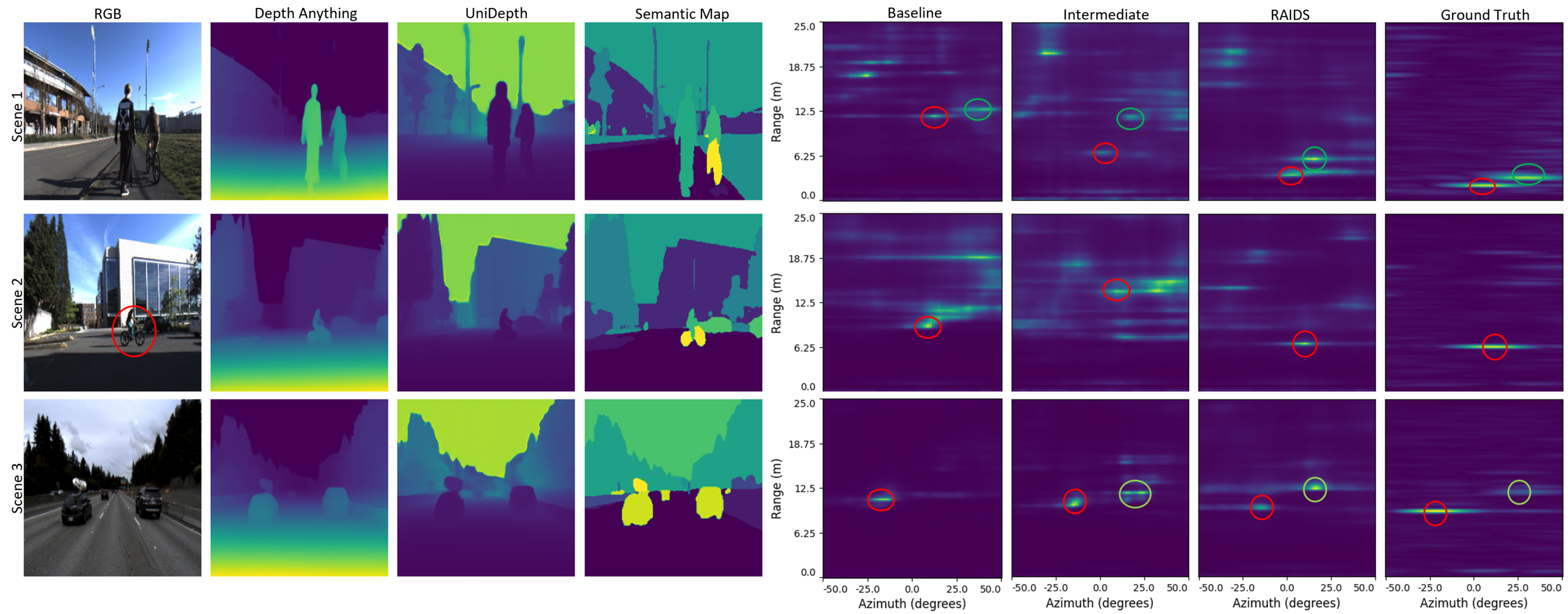


Figure 6.3: Qualitative result on the CRUW dataset.From left to right, each column shows the camera images, estimated depth map from Depth Anything, estimate depth map from UniDepth, semantic segmentation map from Mask2Former, estimated RA maps from the baseline model, RA maps generated by the intermediate model, RA maps generated by RAIDS, and the ground truth RA maps for three different scenes. Every color represents a different object class in the semantic segmentation maps. Key detections are circled in RA maps, and small targets are also circled in the RGB image. Figure best viewed in color.

that is consistent with the ground truth, demonstrating the effectiveness of the CBAM module in enhancing the model's ability to detect and localize targets.

In Fig.6.3 Scene 3, we evaluate the performance of our models on a highway scene that is similar to those found in the RADIal dataset. We observe that all three models detect the cars (activation corresponding to the red circle), but the intermediate and RAIDS models are able to create two distinct activations for the cars (activation corresponding to the red and green circle).

These results demonstrate the ability of the RAIDS to generalize well and process a diverse set of scenarios, including those that contain new semantic classes such as pedestrians or cyclists.

Chapter 7

Conclusions and Future work

In this thesis, we presented the RAIDS model, a novel architecture capable of converting RGB images, supplemented with semantic and depth descriptors, into RA maps using a U-Net (23) inspired CAE architecture with ConvNext (31) blocks. Our proposed models were evaluated on two diverse datasets, RADIaL (33) and CRUW (34), which feature varying environments, including urban, rural, and highway scenes, as well as scenes with pedestrians and cyclists in parking lots. Notably, the two datasets also employ different radar systems with distinct range capabilities, providing a comprehensive test-bed for assessing the generalization capabilities of our models.

Our experimental results demonstrate that the RAIDS model achieves superior performance compared to the baseline model, which relies solely on RGB images as input. This performance difference highlights the limitations of relying solely on RGB data for RA map generation. Furthermore, our ablation study shows that the utilization of partial scene descriptors, including depth and semantic information, leads to significant performance improvements compared to the baseline model. This finding underscores the importance of incorporating scene descriptors into the RA map generation process and supports our hypothesis that explicitly providing depth and contextual information enables our model to focus on synthesizing high-quality RA maps, rather than expending resources on inferring this information.

The RAIDS model also exhibits robust generalization capabilities, performing well on both datasets despite the differences in radar systems and environments. Furthermore, our model demonstrates its ability to generalize to unseen semantics, indicating that it can effectively adapt to new and unfamiliar contexts. This suggests that our model is capable of learning features that are transferable across different domains and semantic scenarios, making it a promising solution for real-world applications where adaptability and flexibility are crucial.

In conclusion, our results provide compelling evidence for the effectiveness of our proposed models in generating accurate RA maps, and highlight the significant benefits of incorporating scene descriptors into the RA map generation process. Our work contributes to the advancement of RA map generation techniques and has potential to impact a wide range of applications, including autonomous driving and surveillance systems.

7.1 Future work

Our work has highlighted the critical role of depth estimation in achieving high-quality RA map synthesis, demonstrating that improved depth estimation can significantly enhance the performance of our RAIDS model. Building on this insight, A promising future direction would be to explore more reliable depth estimation methods, such as utilizing sparse LiDAR point clouds and generative models, such as (9), to convert the sparse point cloud to a dense depth map. This way we can harness the benefits of a more precise source of depth information to further improve the accuracy and realism of our synthesized RA maps. This could involve exploring novel fusion strategies that combine the strengths of our existing model with the advantages of LiDAR based depth estimation.

References

[1] F. Engels, P. Heidenreich, M. Wintermantel, L. Stäcker, M. Al Kadi, and A. M. Zoubir, “Automotive Radar Signal Processing: Research Directions and Practical Challenges,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 15, no. 4, pp. 865–878, 2021. doi: 10.1109/JSTSP.2021.3063666 [Page 1.]

[2] A. Pandharipande, C.-H. Cheng, J. Dauwels, S. Z. Gurbuz, J. Ibanez-Guzman, G. Li, A. Piazzoni, P. Wang, and A. Santra, “Sensing and Machine Learning for Automotive Perception: A Review,” *IEEE Sensors Journal*, vol. 23, no. 11, pp. 11 097–11 115, 2023. doi: 10.1109/JSEN.2023.3262134 [Page 1.]

[3] Y. Wang, Z. Jiang, Y. Li, J.-N. Hwang, G. Xing, and H. Liu, “RODNet: A Real-Time Radar Object Detection Network Cross-Supervised by Camera-Radar Fused Object 3D Localization,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 15, no. 4, pp. 954–967, 2021. doi: 10.1109/JSTSP.2021.3058895 [Page 1.]

[4] T. Alkanat and A. Pandharipande, “Automotive radar point cloud parametric density estimation using camera images,” in *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024. doi: 10.1109/ICASSP48485.2024.10447724 pp. 8636–8640. [Pages 1 and 15.]

[5] J. Yang, B. Ivanovic, O. Litany, X. Weng, S. W. Kim, B. Li, T. Che, D. Xu, S. Fidler, M. Pavone, and Y. Wang, “EmerNeRF: Emergent Spatial-Temporal Scene Decomposition via Self-Supervision,” in *International Conference on Learning Representations*, 2023. [Page 1.]

[6] M. Cordts, M. Omran, S. Ramos, T. Rehfeld, M. Enzweiler, R. Benenson, U. Franke, S. Roth, and B. Schiele, “The Cityscapes Dataset for Semantic Urban Scene Understanding,” in *2016 IEEE*

Conference on Computer Vision and Pattern Recognition (CVPR), 2016. doi: 10.1109/CVPR.2016.350 pp. 3213–3223. [Pages 1 and 20.]

[7] P. Sun, H. Kretzschmar, X. Dotiwalla, A. Chouard, V. Patnaik, P. Tsui, J. Guo, Y. Zhou, Y. Chai, B. Caine, V. Vasudevan, W. Han, J. Ngiam, H. Zhao, A. Timofeev, S. Ettinger, M. Krivokon, A. Gao, A. Joshi, Y. Zhang, J. Shlens, Z. Chen, and D. Anguelov, "Scalability in Perception for Autonomous Driving: Waymo Open Dataset," in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020. doi: 10.1109/CVPR42600.2020.00252 pp. 2443–2451. [Page 1.]

[8] L. Wang, B. Goldluecke, and C. Anklam, "L2r gan: Lidar-to-radar translation," in *Proceedings of the Asian Conference on Computer Vision (ACCV)*, November 2020. [Pages 1 and 14.]

[9] F. Ma and S. Karaman, "Sparse-to-dense: Depth prediction from sparse depth samples and a single image," in *2018 IEEE international conference on robotics and automation (ICRA)*. IEEE, 2018, pp. 4796–4803. [Pages 1 and 42.]

[10] L. Piccinelli, Y.-H. Yang, C. Sakaridis, M. Segu, S. Li, L. Van Gool, and F. Yu, "Unidepth: Universal monocular metric depth estimation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 10 106–10 116. [Pages 3, 19, 30, and 37.]

[11] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. doi: 10.1109/CVPR52688.2022.00135 pp. 1280–1289. [Pages 3, 20, and 30.]

[12] K. K. Patel, "Spatial sparsity based direction of arrival (doa) estimation of acoustic source in the presence of model errors," Ph.D. dissertation, State University of New York at Binghamton, 2019. [Pages ix and 8.]

[13] Z. Wang, A. Bovik, H. Sheikh, and E. Simoncelli, "Image quality assessment: from error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004. doi: 10.1109/TIP.2003.819861 [Page 9.]

[14] V. Mnih, N. Heess, and A. Graves, “Recurrent models of visual attention. advances in neural information processing systems [c],” in *Proc. of Neural Information Processing Systems (NIPS)*, vol. 2, 2014. [Page 10.]

[15] J. Ba, V. Mnih, and K. Kavukcuoglu, “Multiple object recognition with visual attention. arxiv 2014,” *arXiv preprint arXiv:1412.7755*, 2014. [Page 10.]

[16] D. Bahdanau, “Neural machine translation by jointly learning to align and translate,” *arXiv preprint arXiv:1409.0473*, 2014. [Page 10.]

[17] X. Yang, “An overview of the attention mechanisms in computer vision,” in *Journal of physics: Conference series*, vol. 1693, no. 1. IOP Publishing, 2020, p. 012173. [Page 10.]

[18] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, “Cbam: Convolutional block attention module,” in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 3–19. [Pages 10 and 11.]

[19] T. Karras, S. Laine, and T. Aila, “A style-based generator architecture for generative adversarial networks,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 4401–4410. [Page 13.]

[20] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, “Unpaired image-to-image translation using cycle-consistent adversarial networks,” in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 2223–2232. [Page 13.]

[21] M. L. L. de Oliveira and M. J. G. Bekooij, “Generating synthetic short-range fmcw range-doppler maps using generative adversarial networks and deep convolutional autoencoders,” in *2020 IEEE Radar Conference (RadarConf20)*, 2020. doi: 10.1109/RadarConf2043947.2020.9266348 pp. 1–6. [Page 13.]

[22] D. Wachtel, S. Queiroz, T. Schön, W. Huber, and L. Faria, “A novel conditional generative adversarial networks for automotive radar range-doppler targets synthetic generation,” in *2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)*, 2023. doi: 10.1109/ITSC57777.2023.10422067 pp. 3964–3969. [Page 13.]

[23] O. Ronneberger, P. Fischer, and T. Brox, “U-net: Convolutional networks for biomedical image segmentation,” in *Medical image*

computing and computer-assisted intervention–MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III 18. Springer, 2015, pp. 234–241. [Pages 14, 17, 20, and 41.]

[24] A. H. Lang, S. Vora, H. Caesar, L. Zhou, J. Yang, and O. Beijbom, "Pointpillars: Fast encoders for object detection from point clouds," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 12 697–12 705. [Page 14.]

[25] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778. [Page 15.]

[26] Y. Cao, C. Shen, and H. T. Shen, "Exploiting depth from single monocular images for object detection and semantic segmentation," *IEEE Transactions on Image Processing*, vol. 26, no. 2, pp. 836–846, 2016. [Page 17.]

[27] X. Xu, Z. Chen, and F. Yin, "Multi-scale spatial attention-guided monocular depth estimation with semantic enhancement," *IEEE Transactions on Image Processing*, vol. 30, pp. 8811–8822, 2021. [Page 18.]

[28] P. Zama Ramirez, M. Poggi, F. Tosi, S. Mattoccia, and L. Di Stefano, "Geometry meets semantics for semi-supervised monocular depth estimation," in *Computer Vision–ACCV 2018: 14th Asian Conference on Computer Vision, Perth, Australia, December 2–6, 2018, Revised Selected Papers, Part III 14*. Springer, 2019, pp. 298–313. [Page 18.]

[29] L. Yang, B. Kang, Z. Huang, X. Xu, J. Feng, and H. Zhao, "Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data," in *CVPR*, 2024. [Pages 19 and 30.]

[30] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2021, pp. 10 012–10 022. [Page 20.]

[31] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A convnet for the 2020s," in *2022 IEEE/CVF Conference*

on Computer Vision and Pattern Recognition (CVPR), 2022. doi: 10.1109/CVPR52688.2022.01167 pp. 11 966–11 976. [Pages 22, 27, and 41.]

[32] H. Zhao, O. Gallo, I. Frosio, and J. Kautz, “Loss functions for image restoration with neural networks,” *IEEE Transactions on Computational Imaging*, vol. 3, no. 1, pp. 47–57, 2017. doi: 10.1109/TCI.2016.2644865 [Pages 22, 23, and 28.]

[33] J. Rebut, A. Ouaknine, W. Malik, and P. Pérez, “Raw high-definition radar for multi-task learning,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2022, pp. 17 021–17 030. [Pages 27, 28, 30, 31, and 41.]

[34] Y. Wang, Z. Jiang, Y. Li, J.-N. Hwang, G. Xing, and H. Liu, “Rodnet: A real-time radar object detection network cross-supervised by camera-radar fused object 3d localization,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 15, no. 4, pp. 954–967, 2021. [Pages 27, 29, 31, 37, and 41.]

[35] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Köpf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala, “Pytorch: An imperative style, high-performance deep learning library,” 2019. [Online]. Available: https://arxiv.org/abs/1912.01703 [Page 27.]

[36] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *International Conference on Learning Representations*, 2019. [Page 27.]

[37] X. Glorot and Y. Bengio, “Understanding the difficulty of training deep feedforward neural networks,” in *Proceedings of the thirteenth international conference on artificial intelligence and statistics*. JMLR Workshop and Conference Proceedings, 2010, pp. 249–256. [Page 27.]

[38] I. Loshchilov and F. Hutter, “SGDR: Stochastic Gradient Descent with Warm Restarts,” in *5th International Conference on Learning Representations, ICLR 2017, Toulon, France, April 24-26, 2017, Conference Track Proceedings*. OpenReview.net, 2017. [Page 28.]

[39] G. Larsson, M. Maire, and G. Shakhnarovich, “Fractalnet: Ultra-deep neural networks without residuals,” *arXiv preprint arXiv:1605.07648*, 2016. [Page 28.]

[40] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A. C. Berg, and L. Fei-Fei, "ImageNet Large Scale Visual Recognition Challenge," *International Journal of Computer Vision (IJCV)*, vol. 115, no. 3, pp. 211–252, 2015. doi: 10.1007/s11263-015-0816-y [Page 29.]

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